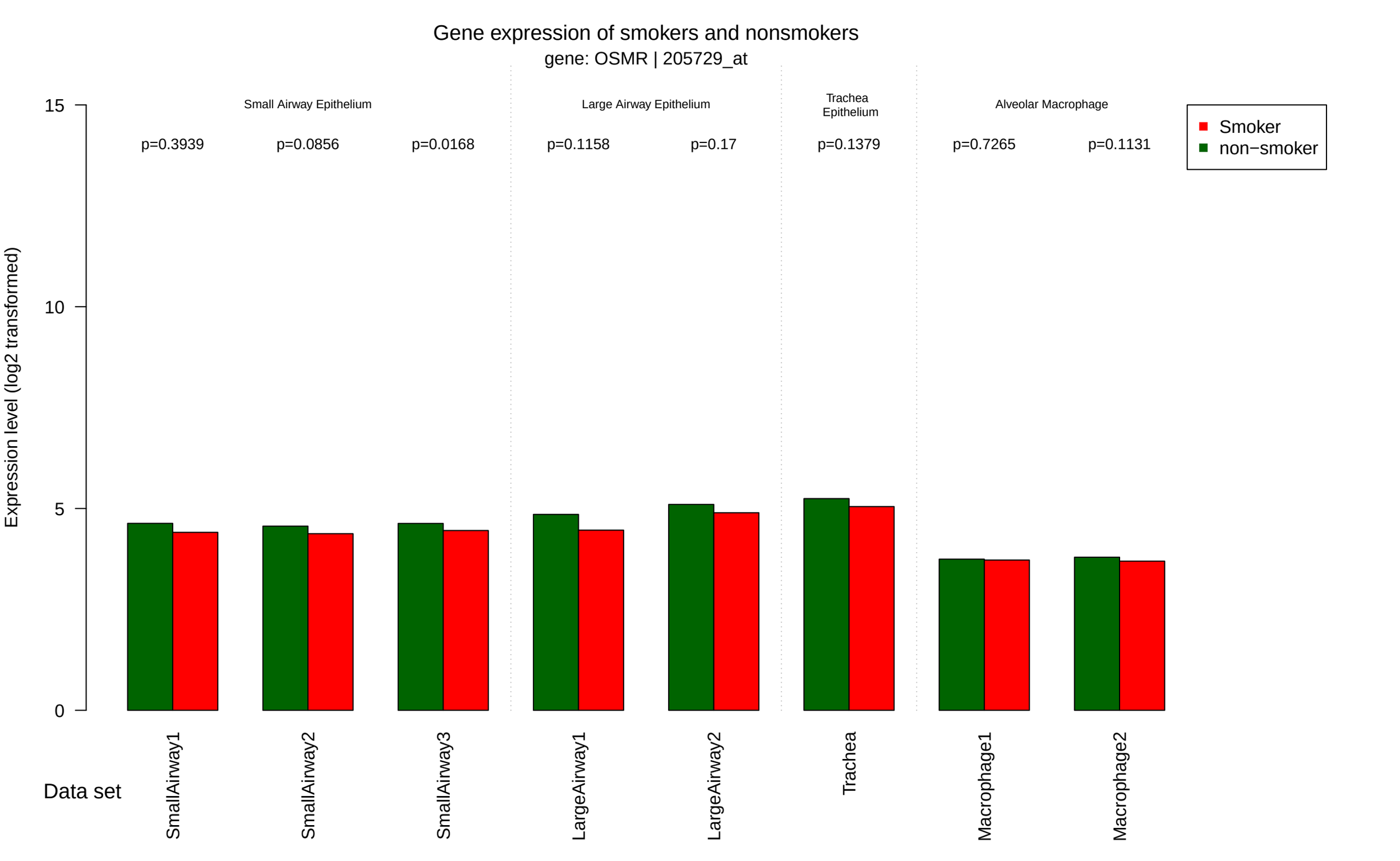
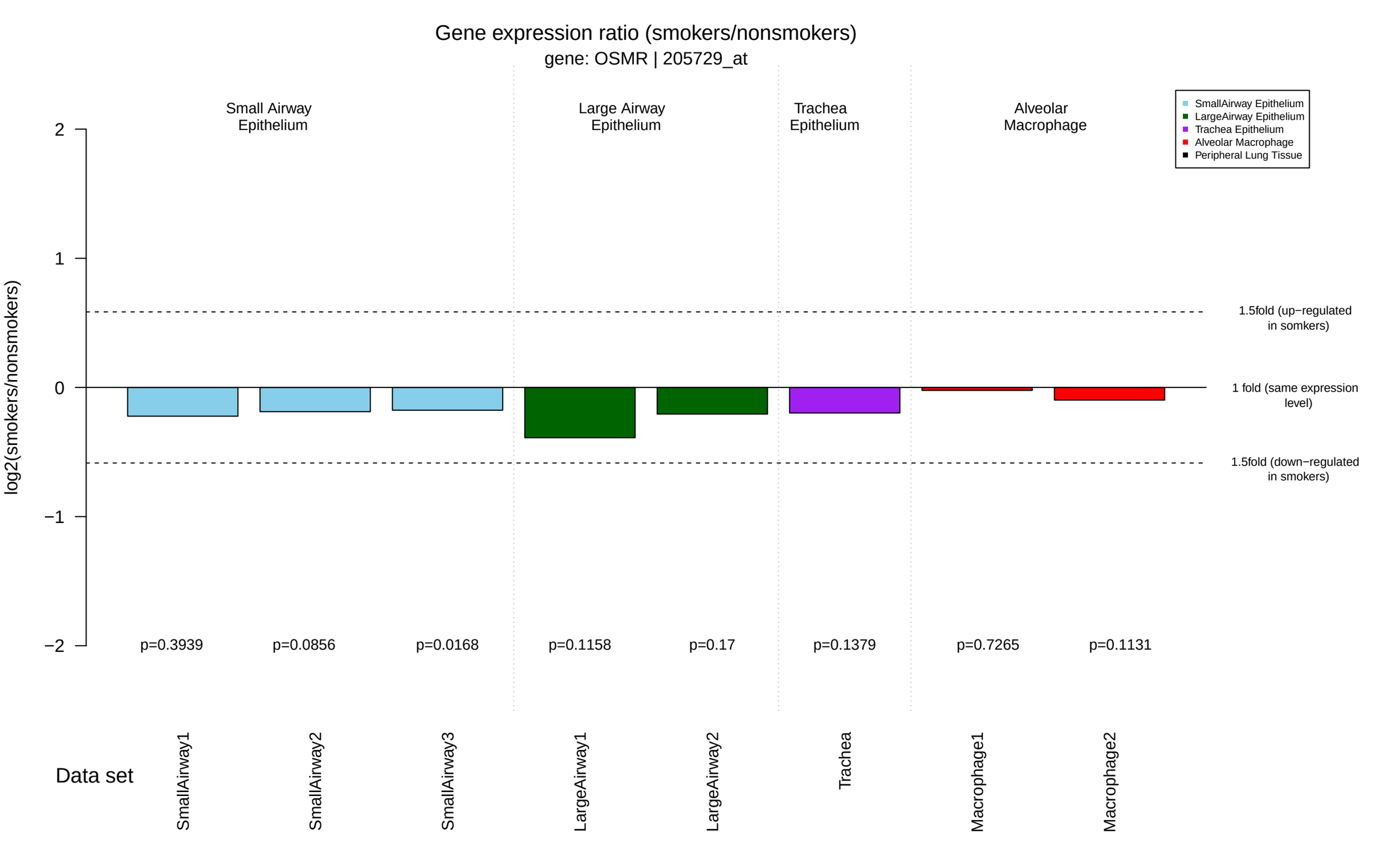
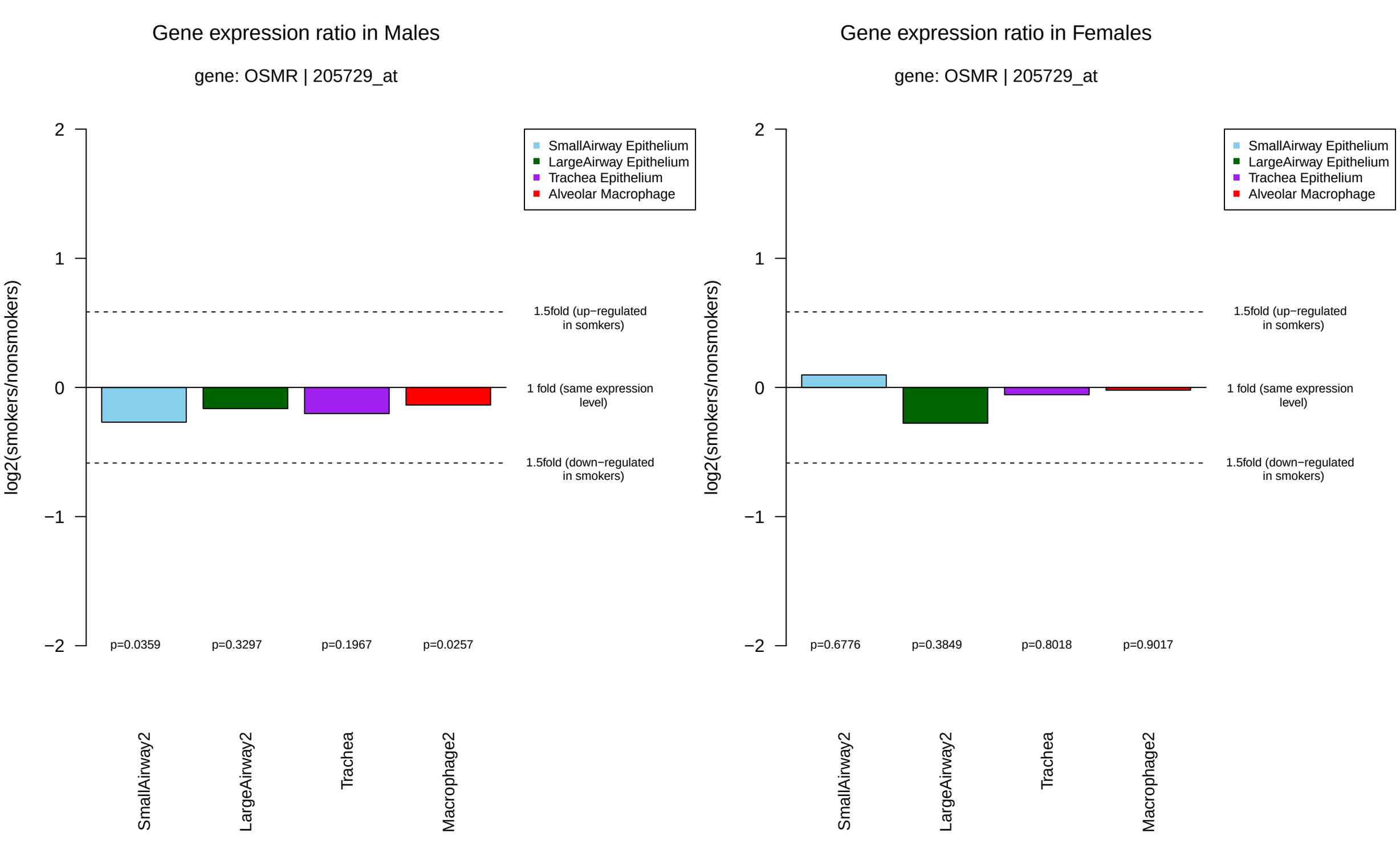
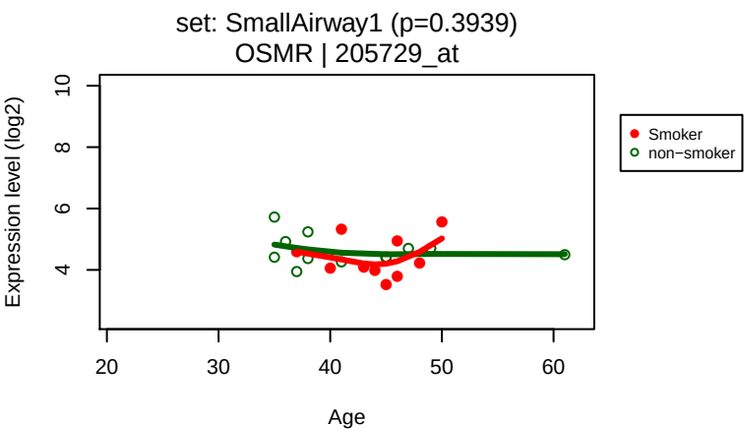
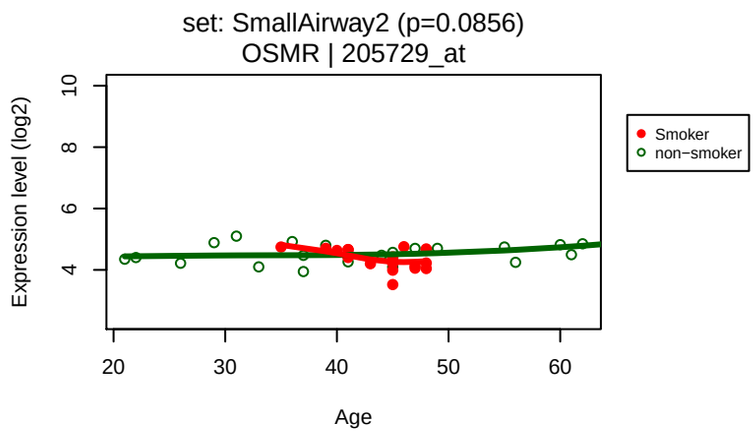
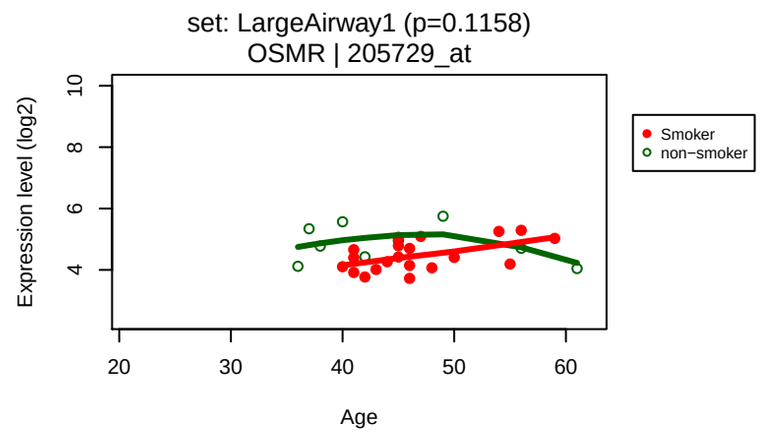

Gene expression of smokers and nonsmokers
gene: OSMR | 205729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3939
p=0.0856
p=0.0168
p=0.1158
p=0.17
p=0.1379
p=0.7265
p=0.1131
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

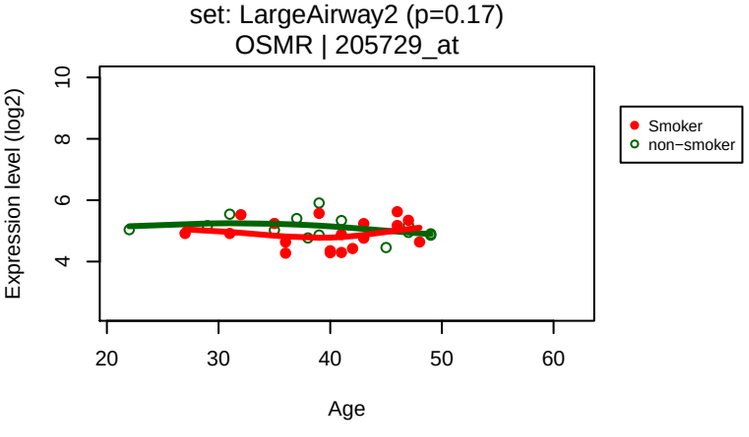

Gene expression ratio (smokers/nonsmokers)
gene: OSMR | 205729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3939
p=0.0856
p=0.0168
p=0.1158
p=0.17
p=0.1379
p=0.7265
p=0.1131
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

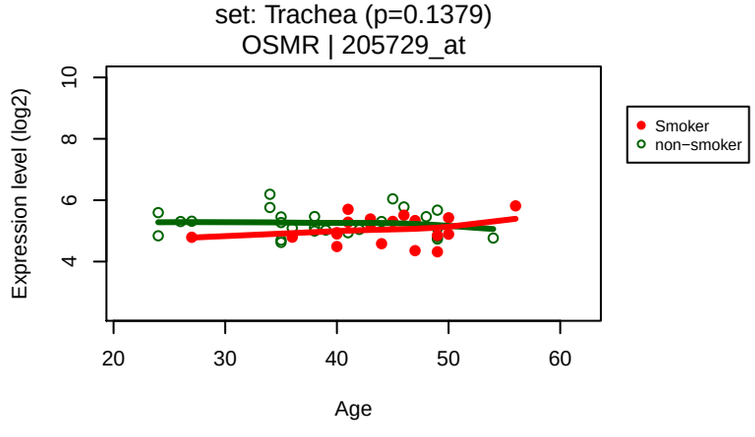

Gene expression ratio in Males
gene: OSMR | 205729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0359
p=0.3297
p=0.1967
p=0.0257
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSMR | 205729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6776
p=0.3849
p=0.8018
p=0.9017
SmallAirway2
LargeAirway2
Trachea
Macrophage2

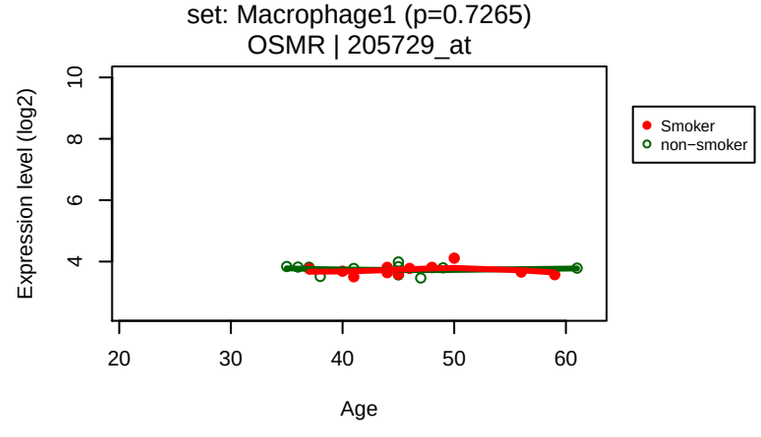

set: SmallAirway1 (p=0.3939)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker

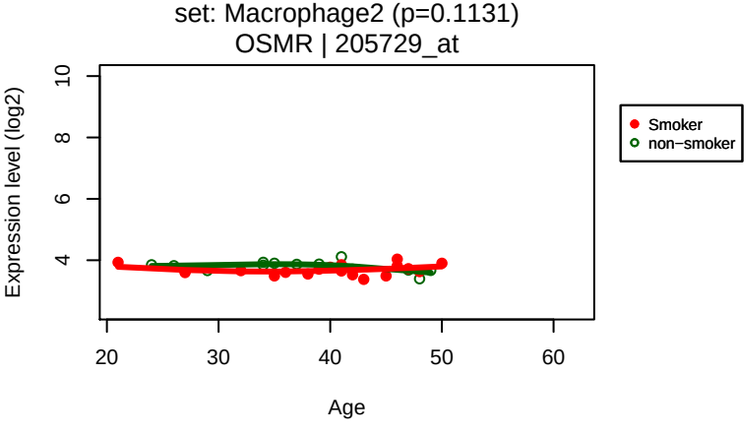

set: SmallAirway2 (p=0.0856)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker

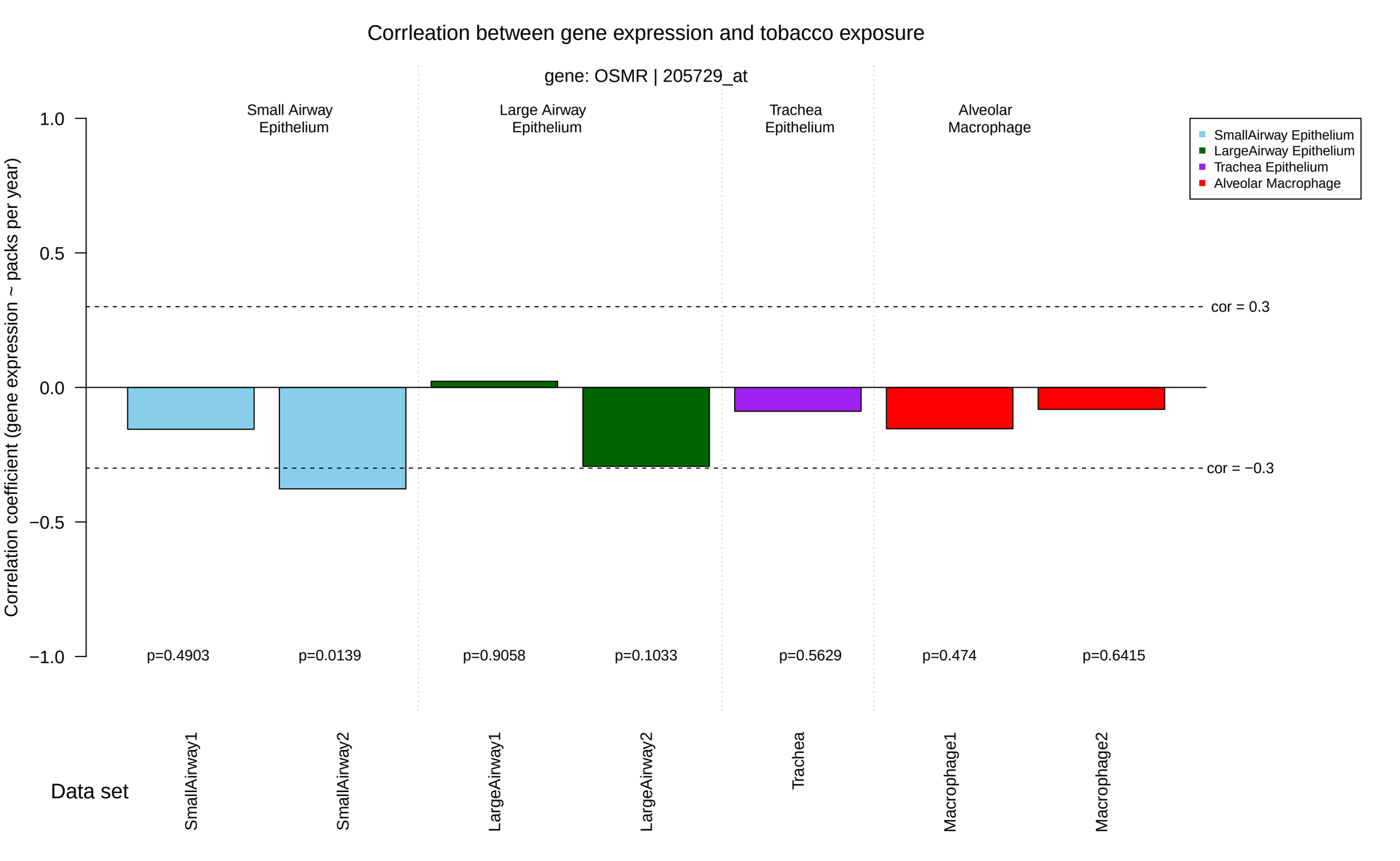

set: LargeAirway1 (p=0.1158)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.17)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1379)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7265)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1131)
OSMR | 205729_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSMR | 205729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4903
p=0.0139
p=0.9058
p=0.1033
p=0.5629
p=0.474
p=0.6415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2